Math 1200 Lecture 8

Sep 29 2026

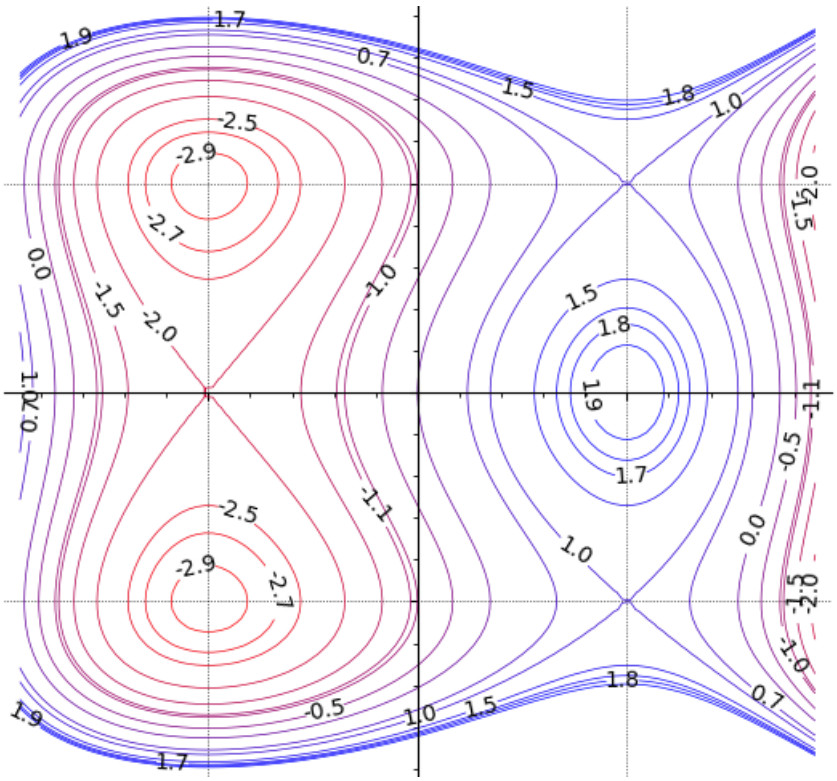
Topics: 14.7: Critical points, second derivative test Name:_____
Collaborators:_____

Group work 1 Find critical points of the following functions

(a) $f(x, y) = \sqrt{x^2 + y^2}$

(b) $f(x, y, z) = xyz$

Group work 2 The following graph is a contour plot of a function $f(x, y)$. Based on the plot find all critical points and identify points as minima, maxima or saddle points.



Group work 3 The plot on the front is of $f(x, y) = 3x - x^3 - 2y^2 + y^4$. Using the second derivative test, verify your classification of critical points.

Group work 4 For which values of a does the function $f(x, y) = x^2 + y^2 + axy$ have a (a) local minimum (b) local maximum (c) saddle point at $(x, y) = (0, 0)$?